Project 02

Laboratórios de Sistemas e Serviços

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Projects

Form groups of two or three students (exceptionally, projects can be done individually) and select **one** of the following projects. Due to all underlying issues with GitHub and GitHub Classroom; all projects will be submitted via **eLearning** in a single compressed file (ZIP), following a standard repository structure.

The ZIP file must contain all relevant scripts, configuration files, a `docker-compose.yml`, and a `README.md` with instructions on how to deploy the project. It should also contain a project report in PDF format.

This is a four-week project (deadline 18/06/2026). You have until the end of this week to notify your professor (via e-mail) of your group members and chosen topic (the list of topics can be found here).

Do not forget to contact your professor with any questions. Further instructions may be added.

Topics

1. IoT Environmental Monitoring System

- **Description:** Simulate multiple IoT sensors (temperature, humidity, air quality) publishing data via **MQTT** to a central broker. A collector service must store this data in a relational database. Use **Grafana** (or a custom frontend) to visualize the historical trends and current status of each sensor. The entire stack must be orchestrated with Docker Compose.
- **Core Topics:** Docker Compose, MQTT (Mosquitto), SQL Persistence, Data Visualization.

2. Real-time Collaborative Task Board

- **Description:** Build a Kanban-style task management application where multiple users can add, move, and edit tasks in real-time. Use **WebSockets** to synchronize the state across all connected clients instantly. All changes must be persisted in a database (SQL or NoSQL) to ensure data is not lost when containers are restarted.
- **Core Topics:** WebSockets, Database Persistence, Frontend Synchronization, Docker Compose.

3. Personal Finance Analytics Platform

- **Description:** Create a RESTful API for tracking personal expenses and income. The system should support categorization and multiple accounts. The frontend must feature interactive charts (e.g., using **Chart.js** or **D3.js**) to visualize spending patterns, monthly budgets, and financial health.
- **Core Topics:** FastAPI/REST, Relational Database (MariaDB/Postgres), Charting/Visualization, Docker Compose.

4. Media Library with Automated Metadata

- **Description:** Build a digital library manager for movies or books. When a user adds a title, the backend fetches rich metadata (posters, ratings, synopses) from an **external REST API** (e.g., TMDB or Google Books). Store the library in a database and provide a visual gallery with filters.
- **Core Topics:** External API Integration, SQL, Multimedia Metadata, Docker.

5. Real-time Online Auction Platform

- **Description:** Implement a bidding system where users can participate in live auctions. Use **WebSockets** to broadcast new bids to all participants immediately. The system must handle bid concurrency using a message broker (like Redis) and persist the final results in a database.
- **Core Topics:** WebSockets, Redis (Pub/Sub), SQL, Atomic Transactions.

6. Geo-Data Dashboard (Traffic or Weather)

- **Description:** Build a web dashboard that visualizes geographical data. You must create a **Python** script that uses an API to get Weather or traffic data or (using **Pandas** or **Polars**) that processes

a dataset (e.g., a CSV of weather stations or traffic incidents with Lat/Lon coordinates) and exports it to JSON. Then, deploy a **Web Server** container (Nginx or Apache) hosting an HTML page. This page must use the **Leaflet** JavaScript library to read that JSON data and display markers on an interactive map.

- **Core Topics:** Web Programming (HTML/JS/Leaflet), Data Formatting (CSV to JSON), Docker, Web Servers.